

torch.linalg.svdvals#

<https://pytorch.org/docs/stable/generated/torch.linalg.svdvals.html>

Description:

Computes the singular values of a matrix. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. The singular values are returned in descending order. This function is equivalent to NumPy's `linalg.svd(A, compute_uv=False)`.

Function Signature

```
torch.linalg.svdvals(A, *, driver=None, out=None) → Tensor#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

`driver` (str, optional) – name of the cuSOLVER method to be used. This keyword argument only works on CUDA inputs. Available options are: `None`, `gesvd`, `gesvdj`, and `gesvda`. Check `torch.linalg.svd()` for details. Default: `None`. `out` (Tensor, optional) – output tensor. Ignored if `None`. Default: `None`.

Returns:

A real-valued tensor, even when `A` is complex.

Code Examples

```
>>> A = torch.randn(5, 3)
>>> S = torch.linalg.svdvals(A)
>>> S
tensor([2.5139, 2.1087, 1.1066])

>>> torch.dist(S, torch.linalg.svd(A, full_matrices=False).S)
tensor(2.4576e-07)
```

Notes & Warnings

NOTE

Note This function is equivalent to NumPy's `linalg.svd(A, compute_uv=False)`.

NOTE

Note When inputs are on a CUDA device, this function synchronizes that device with the CPU.

NOTE

See also `torch.linalg.svd()` computes the full singular value decomposition.

Detailed Documentation

Documentation

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Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

The singular values are returned in descending order.

This function is equivalent to NumPy's `linalg.svd(A, compute_uv=False)`.

When inputs are on a CUDA device, this function synchronizes that device with the CPU.

`torch.linalg.svd()` computes the full singular value decomposition.

A (Tensor) – tensor of shape $(*, m, n)$ where $*$ is zero or more batch dimensions.

`driver` (str, optional) – name of the cuSOLVER method to be used. This keyword argument only works on CUDA inputs. Available options are: `None`, `gesvd`, `gesvdj`, and `gesvda`. Check `torch.linalg.svd()` for details. Default: `None`.

`out` (Tensor, optional) – output tensor. Ignored if `None`. Default: `None`.

A real-valued tensor, even when A is complex.

